RAK13102 WisBlock Blues Notecarrier Module Datasheet

Overview

Description

WisBlock RAK13102 Blues Notecarrier Module is a part of the WisBlock Wireless series. It provides an interface to connect cellular Blues.IO Notecards to the WisBlock system.

Features

- M.2 Blues.IO Notecard connector
- Blues Notecard NOTE-NBGL-500
- Quectel BG-95 M3
- Supports LTE-M, NB-IoT and GSM networks
- IPEX connectors for the cellular and GNSS antenna
- USB Type C debug and log output connector
- Nano SIM and ESIM options
- Two WisBlock Sensor Slot connectors
- 3.3 V power supply (requires an additional battery for operation)
- Small size: 29 mm x 50 mm

Specifications

Overview

The RAK13102 module can be mounted on the IO slot of the WisBlock Base board. **Figure 1** shows the mounting mechanism of the RAK13102 on a WisBlock Base module, such as a RAK19007.

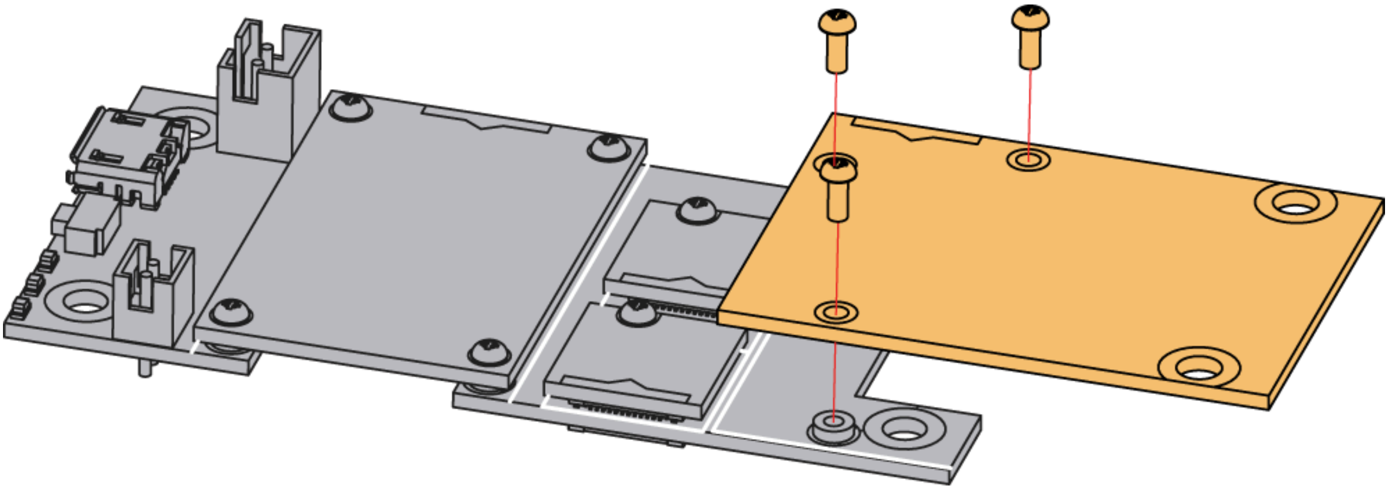

Figure 1: RAK13102 mounting mechanism on a WisBlock Base module

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard ratings of the RAK13102 WisBlock GSM/GPRS Module.

Chipset

Vendor	Part number
Blues.IO	NoteCard NOTE-NBGL-500

Pin Definition

The RAK13102 WisBlock Blues Notecarrier module comprises a standard WisBlock IO slot connector. The WisBlock connector allows the RAK13102 module to be mounted on a WisBlock baseboard with an IO slot, such as RAK19007. The pin order of the connector and the pinout definition are shown in **Figure 2**.

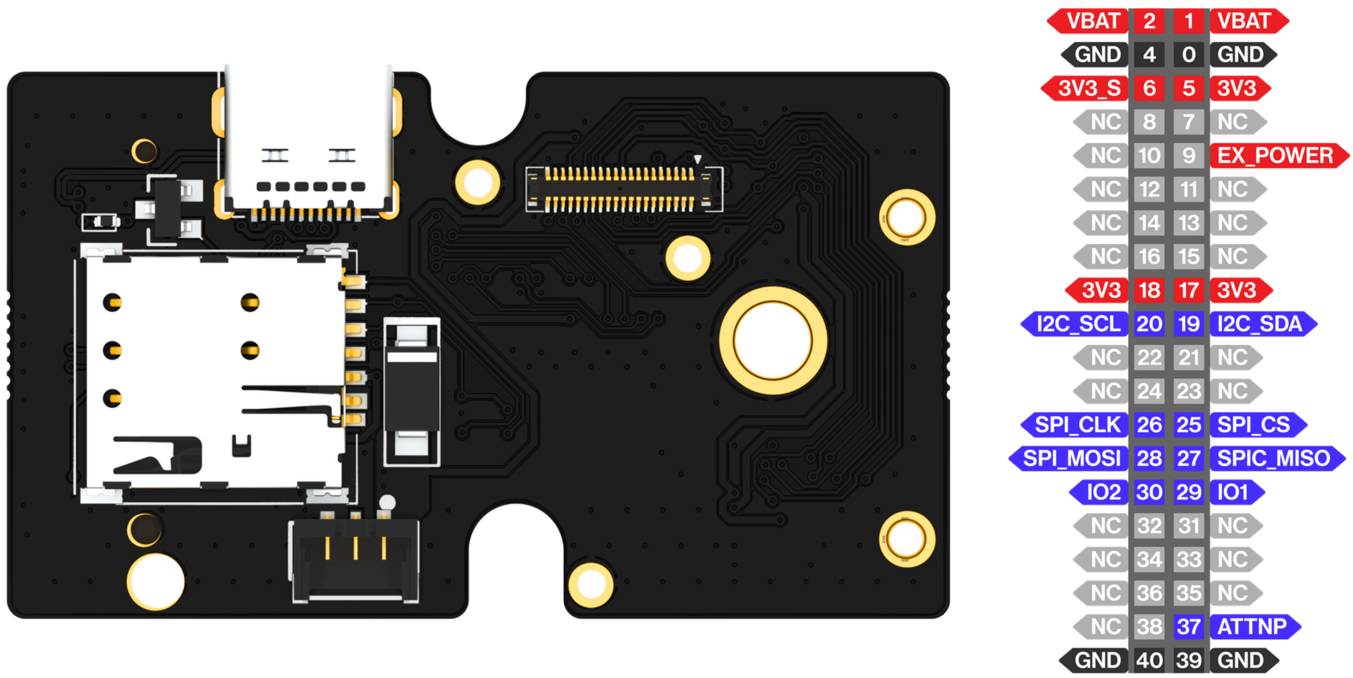

Figure 2: RAK13102 Connector Pin Definition

> **NOTE**
>
> - RAK13102 WisBlock IO slot connector utilizes the **I2C** related pins, **ATTNP** via WB_IO5, **VBAT**, **3V3**, and **GND**.
> - **VBAT** is the battery voltage input with a max voltage of 4.2 V. During cellular data mode, the peak current is ~2000 mA, respectively, which exceeds the USB port supply current. That is why you must use a dedicated battery.

> **IMPORTANT**
>
> This datasheet covers only the specifications of the WisBlock Notecarrier module RAK13102. For detailed information on the Blues.IO Notecard, please check the Blues.IO datasheets from the Blues Resources

Power Supply with External 5V

> **INFO**

The RAK13102 module and the connected WisBlock Base Board and Core module can be supplied with a regulated 5 V DC supply through the P1 connector on the bottom. A matching connector is available with our M8 Power Connector.

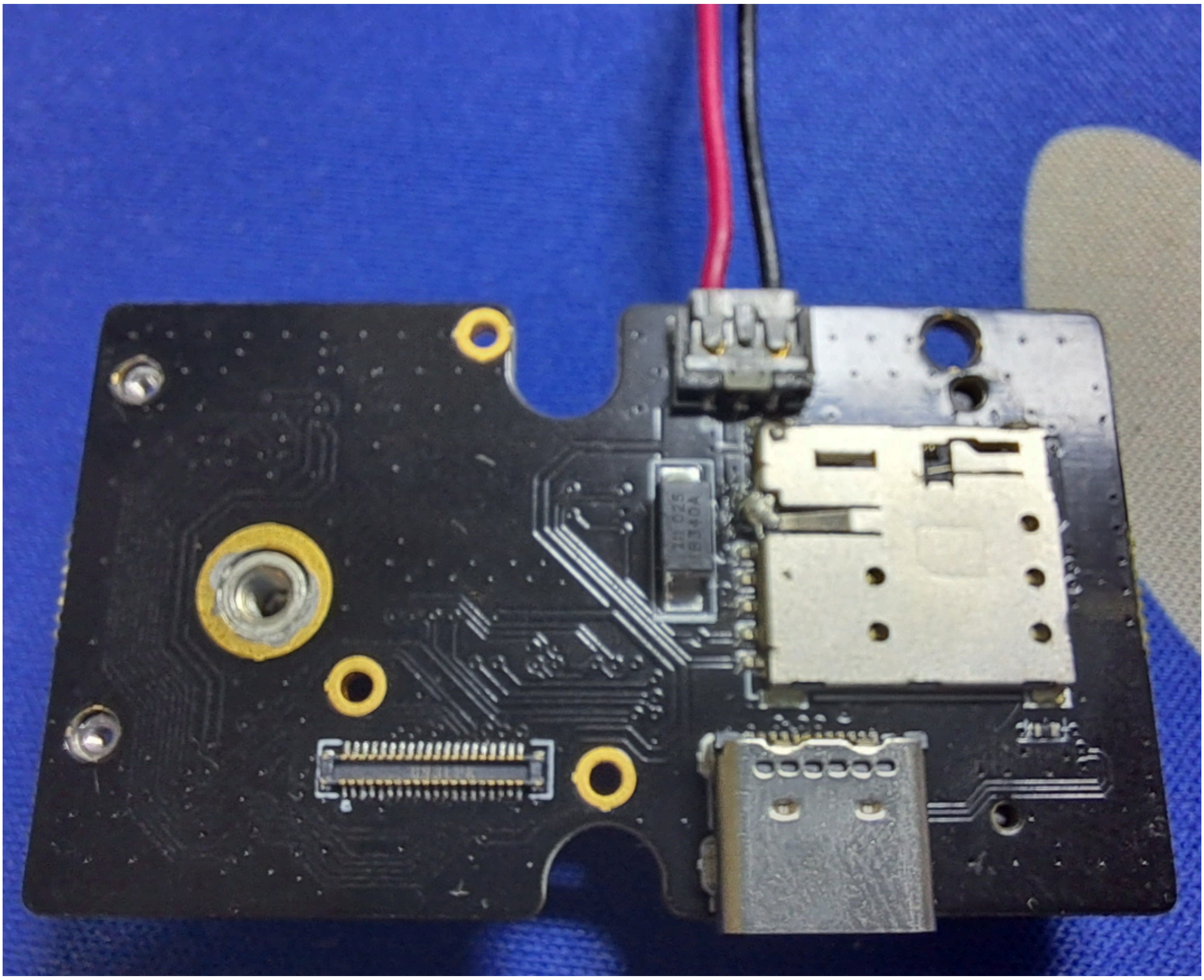

Figure 3: 5V supply through M8 power connector

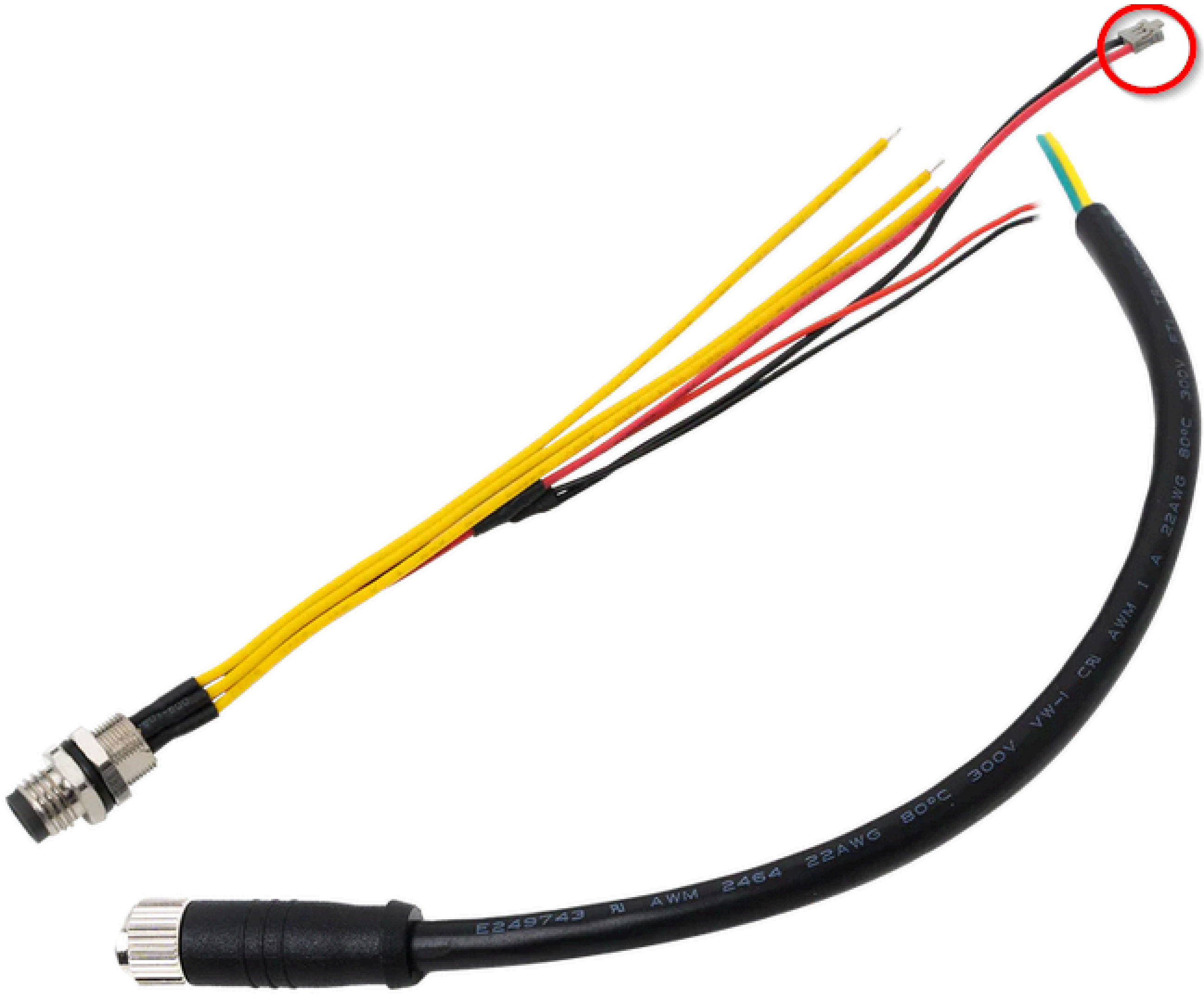

Figure 4: M8 power connector

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit
VBAT	-0.3	+4.2	V
VBUS	-0.3	+5.5	V
EX_POWER	-0.3	+5.5	V

Parameter	Minimum	Maximum	Unit
Power supply peak current0	0	2	A

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
VBAT	Supply Voltage	Input voltage must be within this range	3.3	4.0	4.2	V
EX_POWER	Supply Voltage	Input voltage must be within this range	4.5	5.0	5.5	V

Mechanical Characteristics

Board Dimensions

Figure 5 shows the dimensions and the mechanical drawing of the RAK13102 module.

> **IMPORTANT**
>
> The dimensions are without the Blues Notecard!

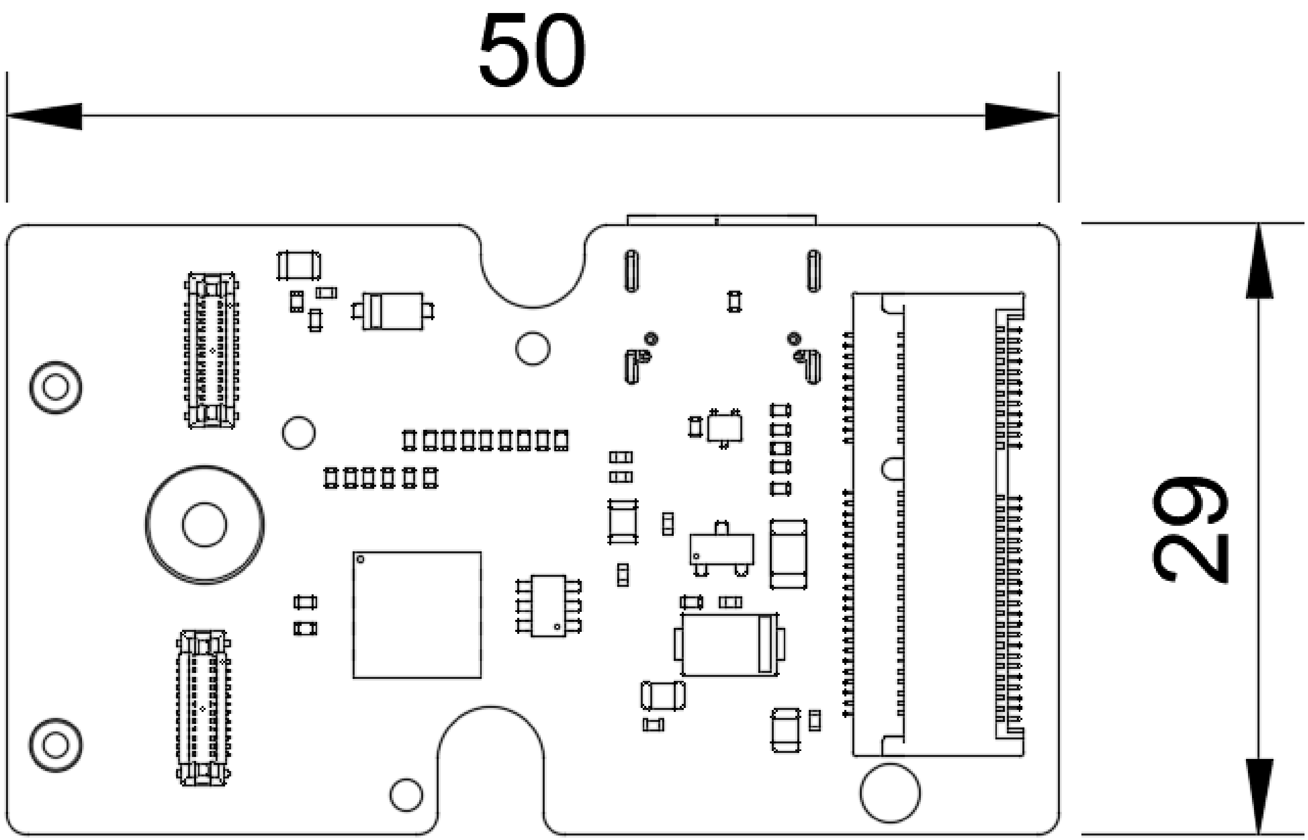

Figure 5: RAK13102 Mechanical Drawing

WisConnector PCB Layout

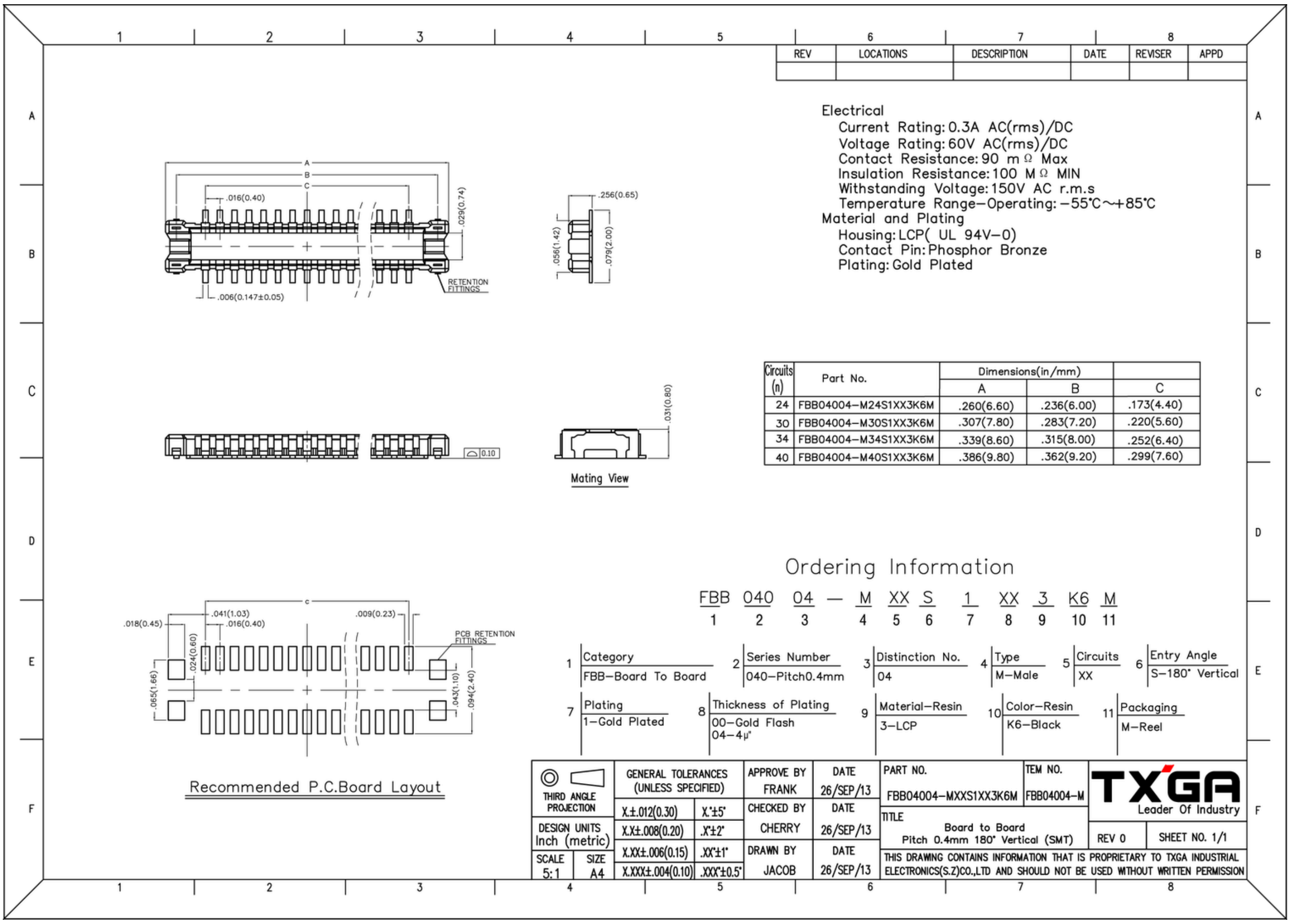

Figure 6: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Power Supply

The Blues.IO Notecard main power supply comes from **VBAT**, which is a battery voltage connected to the WisBlock Base board. **EX_POWER** is the supply from an externally regulated 5 V DC supply.

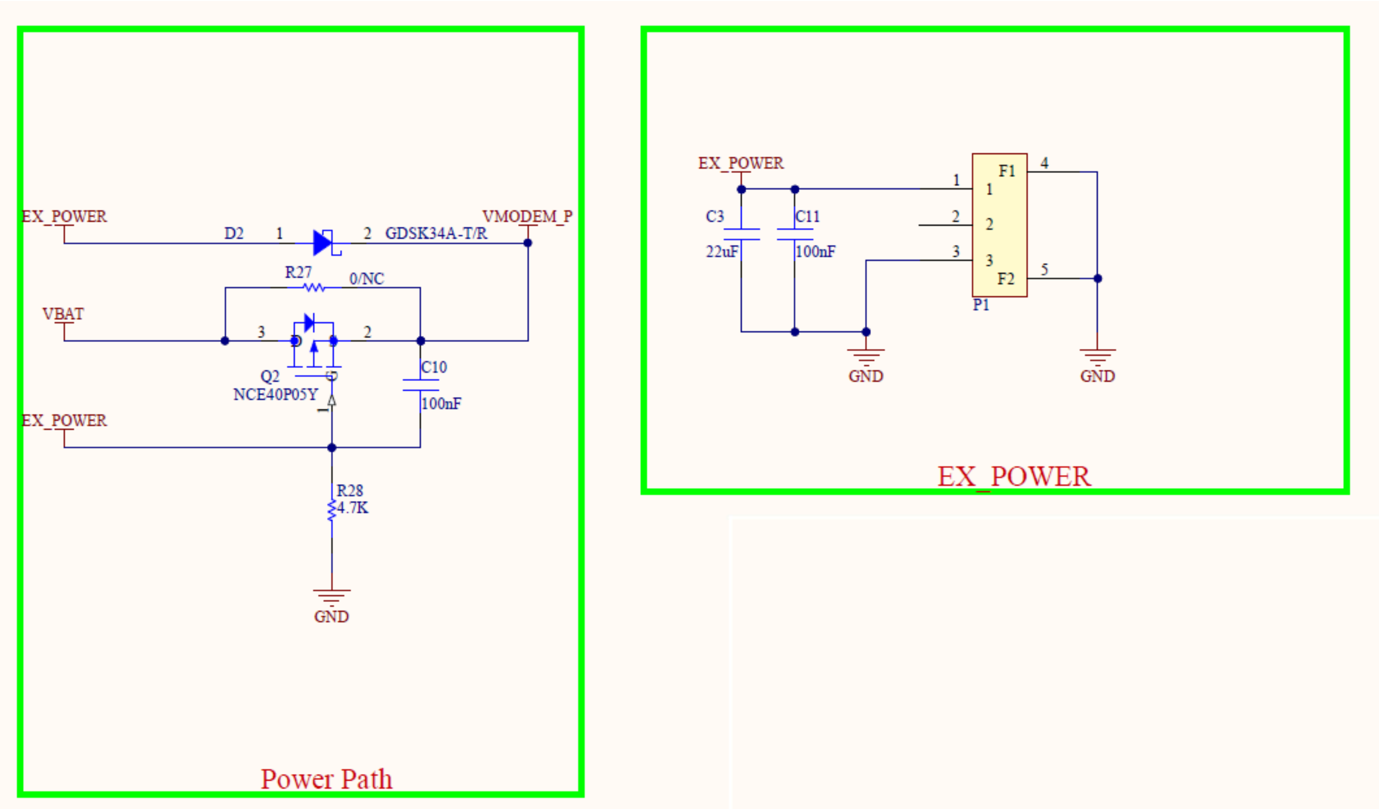

Figure 7: RAK13102 Power Supply

SIM Circuit

Figure 8 shows the schematic of the RAK13102 module with the slot for a standard SIM card and an optional **ESIM** PCB footprint.

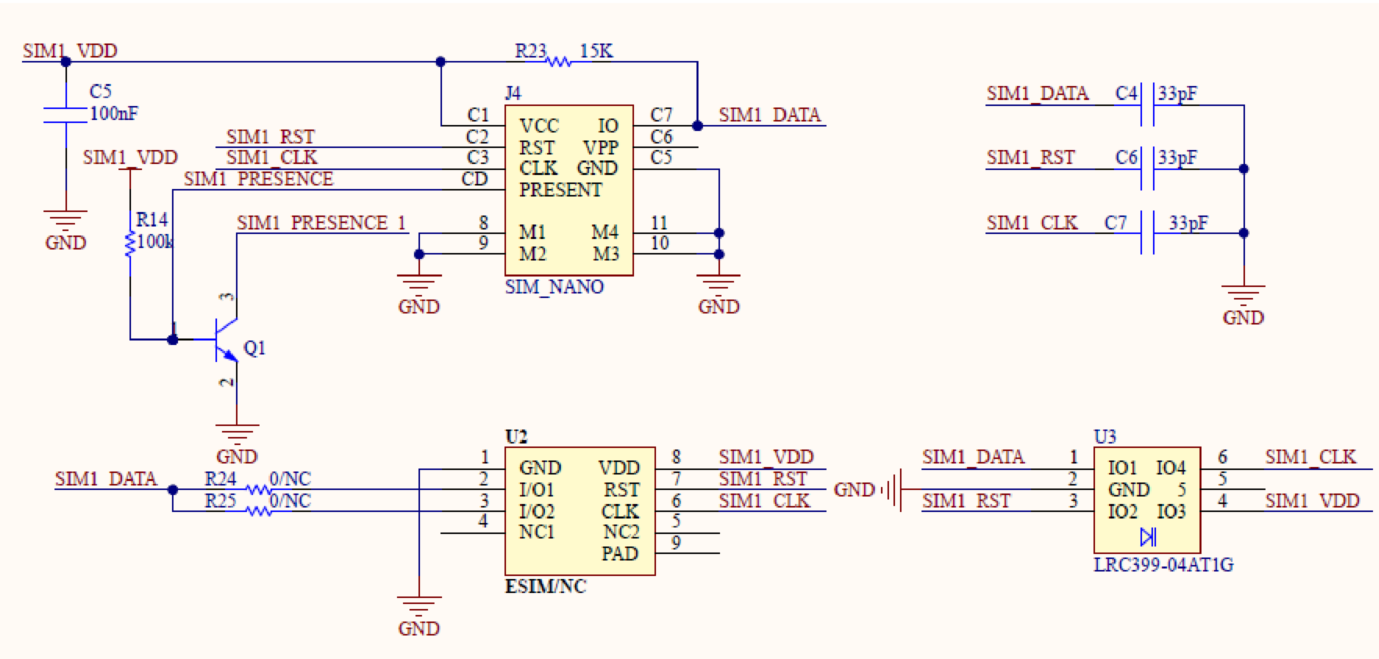

Figure 8: RAK13102 Notecarrier Module SIM Circuit

WisConnector

RAK13102 WisBlock IO slot connector utilizes **I2C** related pins, **ATTNP** via WB_IO5, **VBAT**, **3V3**, and **GND**.

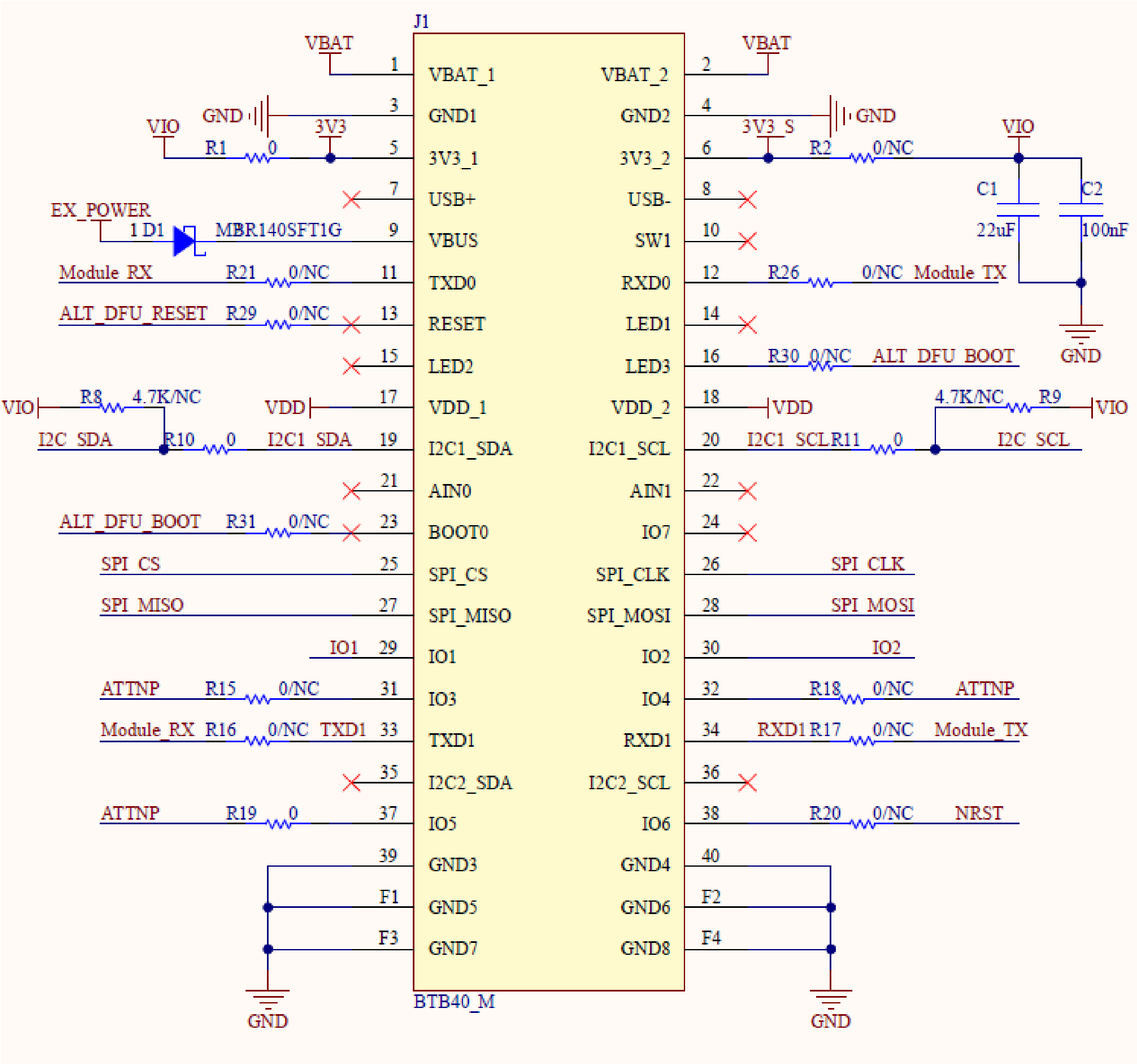

Figure 9: RAK13102 Notecarrier IO Slot Connector

Debug Connector

Figure 10 shows the RAK13102 USB debugging circuit.

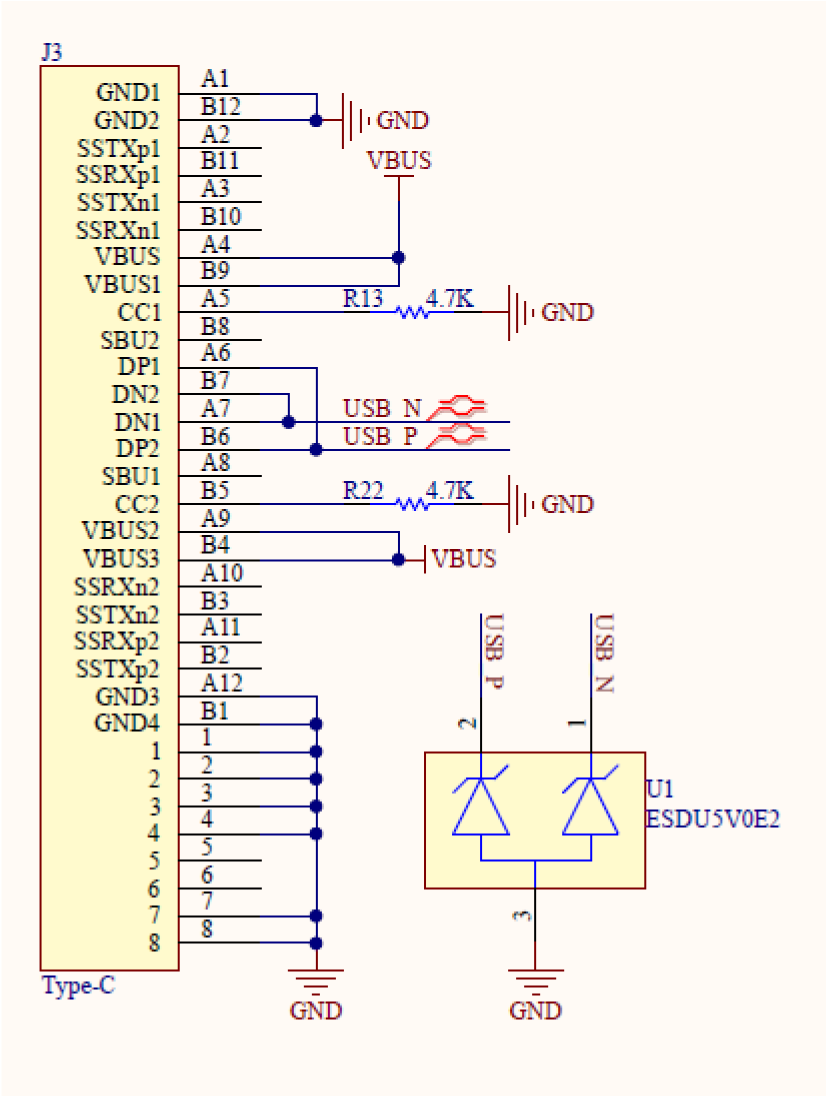

Figure 10: RAK13102 USB Debugging

Blues Notecard M.2 connector

Figure 11 shows the RAK13102 M.2 connector that is the interface to the Blues.IO Notecards.

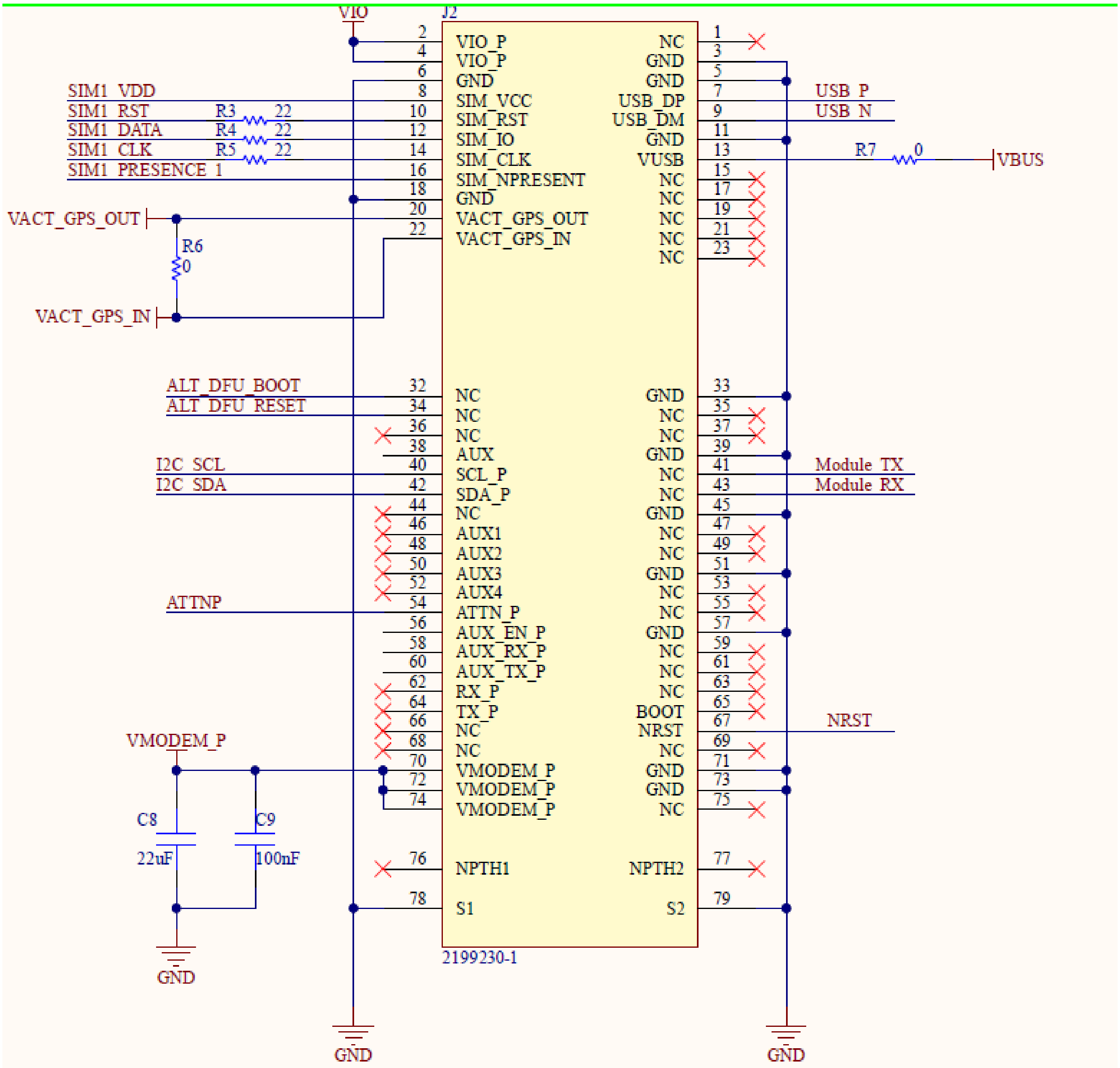

Figure 11: RAK13102 Module M.2 Connector

WisBlock Sensor Slots

NOTE

When assembled in the IO slot, the RAK13102 IO module is covering the standard WisBlock Sensor Slots A and B. These two Sensor Slots are available on the RAK13102 module with the same functionality as the Sensor Slots on the Base Boards

Figure 12 shows the RAK13102 Sensor Slot Connectors.

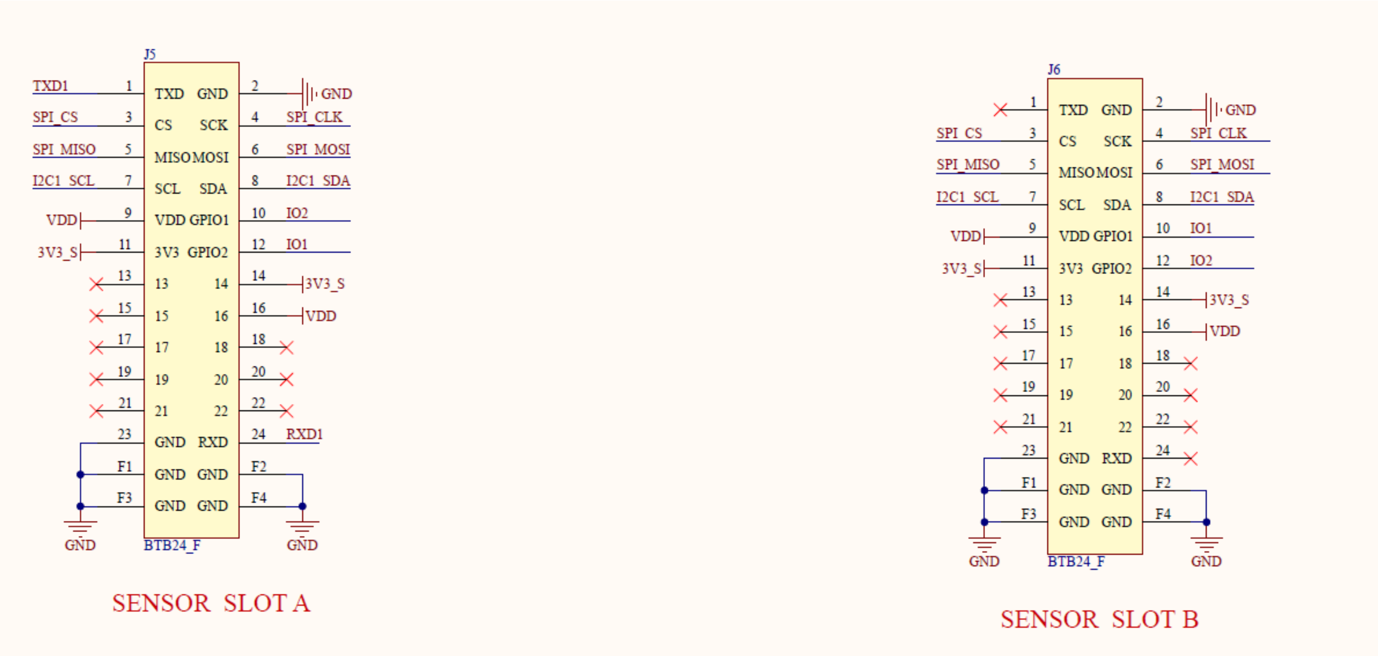

Figure 12: RAK13102 Module Sensor Slots

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